# Week 10

**A company needs to transport goods from three factories (F1, F2, F3) to four warehouses (W1, W2, W3, W4). The supply at each factory and the demand at each warehouse is given in the table below. Use the North-West Corner Rule to find the initial basic feasible solution.**

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To find the initial basic feasible solution using the North-West Corner Rule based on the provided transportation table, we begin at the top-left corner cell (F1, W1) and allocate units step-by-step by comparing the available supply and the corresponding demand.

## Step-by-Step Allocation Process

1. **Cell (F1, W1):**

- **Supply at F1:** 20
- **Demand at W1:** 10
- We allocate the minimum of supply and demand, which is **10**.
- Allocation: $x_{11} = 10$.
- Warehouse W1 demand is fully satisfied (Demand becomes 0). Factory F1 remaining supply is $20 - 10 = 10$. Since W1 demand is exhausted, we move horizontally to the next column (W2).

2. **Cell (F1, W2):**

- **Remaining Supply at F1:** 10
- **Demand at W2:** 25
- We allocate the minimum of remaining supply and demand, which is **10**.
- Allocation: $x_{12} = 10$.
- Factory F1 supply is now fully exhausted (Supply becomes 0). Warehouse W2 remaining demand is $25 - 10 = 15$. Since F1 supply is exhausted, we move vertically down to the next row (F2).

3. **Cell (F2, W2):**

- **Supply at F2:** 30
- **Remaining Demand at W2:** 15
- We allocate the minimum of supply and remaining demand, which is **15**.
- Allocation: $x_{22} = 15$.
- Warehouse W2 demand is fully satisfied (Demand becomes 0). Factory F2 remaining supply is $30 - 15 = 15$. Since W2 demand is exhausted, we move horizontally to the next column (W3).

4. **Cell (F2, W3):**

- **Remaining Supply at F2:** 15
- **Demand at W3:** 20
- We allocate the minimum of remaining supply and demand, which is **15**.
- Allocation: $x_{23} = 15$.
- Factory F2 supply is now fully exhausted (Supply becomes 0). Warehouse W3 remaining demand is $20 - 15 = 5$. Since F2 supply is exhausted, we move vertically down to the next row (F3).

**5. Cell (F3, W3):**

- **Supply at F3:** 25
- **Remaining Demand at W3:** 5
- We allocate the minimum of supply and remaining demand, which is **5**.
- Allocation: $x_{33} = 5$.
- Warehouse W3 demand is fully satisfied (Demand becomes 0). Factory F3 remaining supply is $25 - 5 = 20$. Since W3 demand is exhausted, we move horizontally to the next column (W4).

**6. Cell (F3, W4):**

- **Remaining Supply at F3:** 20
- **Demand at W4:** 20
- We allocate the minimum of remaining supply and demand, which is **20**.
- Allocation: $x_{34} = 20$.
- Both F3 supply and W4 demand are fully satisfied (both become 0).

## Summary of Allocations (Initial Basic Feasible Solution)

- $x_{11} = 10$ (Cost: $6 \times 10 = 60$)
- $x_{12} = 10$ (Cost: $4 \times 10 = 40$)
- $x_{22} = 15$ (Cost: $8 \times 15 = 120$)
- $x_{23} = 15$ (Cost: $7 \times 15 = 105$)
- $x_{33} = 5$ (Cost: $5 \times 5 = 25$)
- $x_{34} = 20$ (Cost: $3 \times 20 = 60$)

## Total Transportation Cost Calculation

Total Cost = $(10 \times 6) + (10 \times 4) + (15 \times 8) + (15 \times 7) + (5 \times 5) + (20 \times 3)$

Total Cost = $60 + 40 + 120 + 105 + 25 + 60 = 410$